

AdVantage AI

System Models Specification & Data Dictionary

Executive Summary

AdVantage AI is a full-stack AI Marketing Campaign Management System. This document defines the complete model architecture powering the system, spanning **Database Data Models (Mongoose ORM)**, **External AI Generation Models (Pollinations LLM & Diffusion)**, and **System Functional Modules**.

1. Database Schemas & Backend Models (MongoDB / Mongoose)

The backend database layer consists of 5 core Mongoose collections stored in MongoDB. Below are the precise field definitions, data types, and operational requirements for each model.

1.1 Campaign Model (backend/models/Campaign.js)

Field Name	Type	Required	Default	Description
name	String	Yes	—	Title of the marketing campaign
objective	String	Yes	—	Campaign goal (e.g. Lead Gen, Conversion, Awareness)
status	String	No	'Active'	Operational state (Active, Paused, Completed)
budget	Number	Yes	—	Total budget allocation amount (\$)
roi	Number	No	0	Return on Investment ratio multiplier
startDate	String	Yes	—	Campaign launch date (YYYY-MM-DD)
endDate	String	Yes	—	Campaign conclusion date (YYYY-MM-DD)
channels	Array [String]	No	[]	Target platforms (Instagram, LinkedIn, Twitter, etc.)

1.2 Post Model (backend/models/Post.js)

Field Name	Type	Required	Default	Description
title	String	Yes	—	Post headline or product summary topic
platform	String	Yes	—	Social platform (Instagram, LinkedIn, Twitter, Facebook)
content	String	Yes	—	Generated copy caption, hashtags, and formatting
imageUrl	String	No	"	Pollinations AI generated image banner URL
publishedAt	Date	No	Date.now	Timestamp when post was recorded

1.3 Profile Model (backend/models/Profile.js)

Field Name	Type	Required	Default	Description
name	String	Yes	—	Full name of account manager
email	String	Yes	—	User work email address
company	String	Yes	—	Associated company or brand organization
phone	String	No	"	Contact phone number
industry	String	No	"	Business vertical or industry domain
avatar	String	No	"	Base64 encoded string or profile image URL

1.4 Notification Model (backend/models/Notification.js)

Field Name	Type	Required	Default	Description
title	String	Yes	—	Notification header summary
text	String	Yes	—	Detailed message alert content
type	String	No	'info'	Alert visual category ('success', 'info', 'error')
unread	Boolean	No	true	Status indicator for unread alert badge

1.5 Audit / History Model (backend/models/History.js)

Field Name	Type	Required	Default	Description
action	String	Yes	—	Logged system operation title
category	String	Yes	—	Category ('campaign', 'post', 'profile', 'email', 'ai-content')
details	String	Yes	—	Full descriptive audit context and metadata
createdAt	Date	No	Date.now	System timestamp of logged event (Mongoose auto)

2. AI & Generation Models Architecture

AdVantage AI integrates external generative models alongside high-availability local fallback logic to deliver automated copywriting and dynamic image asset creation.

Model Component	Endpoint / Provider	Input Prompt / Trigger	Output Generated
Pollinations Text LLM Model	https://text.pollinations.ai/	Product summary string + target platform constraints	JSON object containing Instagram, LinkedIn, Twitter, Facebook captions & HTML Email body

Pollinations Image Diffusion Model	https://image.pollinations.ai/prompt/{prompt}	Encoded product context prompt with negative keywords (no text/logos)	High-resolution 800x600 product image visual link
Rule-Based Local Fallback Model	Local Frontend Execution (ContentPreview.jsx)	Network disconnect or external API timeout trigger	Pre-compiled structured template copy for all 5 target platforms

3. System Functional Modules Mapping

The table below illustrates how front-end React components map to backend Mongoose data models and external AI endpoints across the application.

Module Name	React Component	Associated Backend Model	Key Functional Capabilities
Dashboard Control Room	Dashboard.jsx	Campaign, History, Notification	Displays active campaign count, total budget, ROI stats, and live audit feed.
Campaign Manager	Campaigns.jsx	Campaign Model	Full CRUD operations for campaign tracking, schedule dates, and budget allocations.
AI Post Creator	ContentPreview.jsx	Post Model + Pollinations AI	Generates 5-channel copy, renders live smartphone UI mockups (IG, LinkedIn, Twitter, FB, Email).
Email Creator	EmailCreator.jsx	History Model	Compiles HTML email templates (Promotional, Onboarding, Launch) with live sandbox preview.
Analytics Board	Analytics.jsx	Campaign Model	Renders Chart.js visualizations for engagement trends, channel split, and budget vs ROI ratios.
Audit & History Log	HistoryLog.jsx	History Model	Filterable historical record of all campaign additions, AI generation events, and profile edits.
Profile & Settings	Settings.jsx	Profile Model	Manages corporate metadata, industry selection, contact info, and Base64 avatar sync.